

INNOVATION PROJECT REPORT

CHE212

Experimental Study of Heat Loss Variation with Insulation Thickness and Critical Radius in a Cylindrical System

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1 Abstract

This study investigates the variation of heat loss from a cylindrical system with increasing insulation thickness. Unlike flat surfaces, insulation in cylindrical geometries exhibits a non-monotonic behavior where heat loss initially increases and then decreases. This phenomenon arises due to the competing effects of increasing convective surface area and increasing conduction resistance.

The experiment was performed using an MS cylinder with externally applied silicone rubber insulation of varying thickness. Heat input was measured using electrical power, and corresponding heat loss was analyzed for different outer radii.

The results show that heat loss increases initially, reaches a maximum value at a certain radius, and then decreases with further insulation. This maximum point corresponds to the critical radius of insulation. Experimental results were compared with theoretical predictions using thermal resistance models and Nusselt number correlations, showing good agreement.

2 Objective

- To study variation of heat loss with insulation thickness
- To determine critical radius experimentally
- To compare theoretical and experimental results
- To evaluate heat transfer coefficient using different approaches

3 Theory

3.1 Modes of Heat Transfer

Heat transfer from the cylindrical system occurs through:

- Conduction through insulation
- Convection from outer surface

3.2 Radial Conduction in Cylinders

From Fourier's law in cylindrical coordinates:

$$Q = -kA \frac{dT}{dr} \quad (1)$$

For steady-state radial conduction:

$$Q = \frac{2\pi kL(T_1 - T_2)}{\ln(r_2/r_1)} \quad (2)$$

Observation: Resistance increases logarithmically with radius.

3.3 Convective Heat Transfer

$$Q = hA(T_s - T_\infty) \quad (3)$$

Where:

$$A = 2\pi r_2 L \quad (4)$$

3.4 Thermal Resistance Network

Total resistance is:

$$R_{cond} = \frac{\ln(r_2/r_1)}{2\pi kL} \quad (5)$$

$$R_{conv} = \frac{1}{2\pi r_2 Lh} \quad (6)$$

$$R_{total} = R_{cond} + R_{conv} \quad (7)$$

$$Q = \frac{T_1 - T_\infty}{R_{total}} \quad (8)$$

3.5 Critical Radius of Insulation (Detailed Derivation)

Heat transfer expression:

$$Q = \frac{T_1 - T_\infty}{\frac{\ln(r_2/r_1)}{2\pi kL} + \frac{1}{2\pi r_2 Lh}} \quad (9)$$

To find maximum heat loss:

$$\frac{dQ}{dr_2} = 0 \quad (10)$$

Solving:

$$r_c = \frac{k}{h} \quad (11)$$

Physical Meaning:

- Increase in r_2 increases convection area
 - Increase in r_2 increases conduction resistance
- Thus:

- For $r_2 < r_c$: Heat loss increases
- For $r_2 > r_c$: Heat loss decreases

3.6 Convective Heat Transfer Coefficient

3.6.1 Using Nusselt Number

$$Nu = \frac{hD}{k_{air}} \quad (12)$$

Churchill-Chu correlation:

$$Nu = \left[0.60 + \frac{0.387Ra^{1/6}}{(1 + (0.559/Pr)^{9/16})^{8/27}} \right]^2 \quad (13)$$

3.6.2 Using Experimental Heat Loss

$$Q_{total} = V I \quad (14)$$

$$Q_{final} = Q_{total} - Q_{rad} - Q_{ends} \quad (15)$$

$$h = \frac{Q_{final}}{2\pi r L (T_s - T_\infty)} \quad (16)$$

4 Experimental Setup

- MS cylinder (Length = 10 cm, Radius = 12 mm)
- Silicone rubber insulation ($k \approx 0.25$ W/mK)
- Heater (7V supply)
- Thermocouples

5 Methodology

1. Perform experiment without insulation
2. Add insulation in layers (4 mm)
3. Apply constant voltage
4. Wait for steady state
5. Record V , I , and temperatures
6. Repeat for different radii

6 Experimental Data

6.1 Raw Data

Radius (mm)	Voltage (V)	Current (A)	Heat Loss (W)
12	7	1.87	13.09
16	7	1.90	13.30
20	7	1.90	13.30
24	7	1.89	13.23
28	7	1.89	13.23
32	7	1.87	13.09

6.2 Corrected Data

r (mm)	Q_{total}	Q_{rad}	Q_{ends}	Q_{final}	h
12	13.09	1.40	1.57	10.12	64.2
16	13.30	1.07	2.11	10.12	61.9
20	13.30	1.03	3.30	8.97	58.0
24	13.23	0.99	4.03	8.21	52.9

7 Results and Discussion

- Heat loss increases initially due to increased surface area
- Maximum heat loss observed around 20–24 mm
- After this, heat loss decreases due to conduction resistance

(17)

$$r_c \approx 21.5\text{mm}$$

8 Cooling Behavior

Cooling follows Newton's law:

$$T(t) = T_{\infty} + (T_0 - T_{\infty})e^{-kt} \quad (18)$$

9 Sources of Error

- Radiation estimation error
- Contact resistance
- Temperature measurement error
- Non-uniform insulation

10 Applications

- Steam pipelines
- Heating systems
- Electrical insulation

- Industrial heat transfer systems

11 Conclusion

- Heat loss initially increases then decreases
- Critical radius verified experimentally
- Experimental results match theory